JOURNAL OF INDONESIA SCIENTIFIC SOCIETY

Published by Indonesia Scientific Society (ISS) • ISSN 2775-365X • https://journal-iss.org

Vol. 01, No. 02, November, pp. 12–26

Research Article

Topology Optimization as a Strategy for Mitigating Stress Shielding in Orthopedic Bone Implants

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Article Information

Received: August 2026
Revised: September 2026
Accepted: Oktober 2026
Published: November 2026

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Citation: Ahmad, M. R., & Gupta, A. (2026). Topology optimization as a strategy for mitigating stress shielding in orthopedic bone implants. *Journal of Indonesia Scientific Society*.

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Abstract

Orthopedic implants made from titanium alloy Ti-6Al-4V have an elastic modulus of approximately 110 GPa, nearly six times the stiffness of cortical bone. This mismatch produces stress shielding, where the rigid implant absorbs most of the mechanical load, and the surrounding bone receives insufficient stimulation to remain healthy. Under Wolff's Law, chronically under-loaded bone is progressively resorbed and weakened, leading to implant loosening and revision surgery. This study applied the Solid Isotropic Material with Penalization (SIMP) topology optimization algorithm, widely used in aerospace structural design, to a two-dimensional finite element model of a titanium bone fixation plate implemented from scratch in Python. A parametric study was conducted across five volume fractions from 30% to 70%, and relative effective stiffness of each optimized topology were evaluated. Additional investigations included mesh-convergence analysis, bending-versus-axial load case comparison and computational load response validation. None of the five volume fractions produced an implant with effective stiffness inside the cortical bone range. The closest result was at 30% volume fraction, which achieved an effective stiffness of 37.2 GPa, a 66.2% reduction from the 110 GPa titanium baseline, narrowing the absolute stiffness gap from 81 to 89 GPa down to 8 to 16 GPa. These findings suggest that topology optimization is a viable approach for mitigating stress shielding in orthopedic implants, with further stiffness reduction possible through lower volume fractions or alternative lower-modulus implant materials.

Keywords: Topology Optimization; Stress Shielding; Bone Implant Design; SIMP Algorithm; Finite Element Method

1. Introduction

7.5 million orthopedic devices are implanted every year in the United States alone, with the global market for them estimated to reach $79.5 billion by 2030 [13]. These devices are designed to stabilize damaged bone and restore normal load-bearing function. While short-term surgical outcomes are generally positive, long-term implant failure remains a significant clinical problem.

One of the most well-documented causes of long-term failure is stress shielding [5]. Under Wolff's Law, when bone tissue experiences a sustained reduction in mechanical load, the internal structure of trabecular bone becomes less dense and less stiff over time, weakening the bone's ability to carry load [4].

Stress shielding occurs because most orthopedic implants are made from metals such as titanium alloy Ti-6Al-4V, which has an elastic modulus of approximately 110 GPa [6]. Cortical bone, by comparison, has an elastic modulus of only 21 to 29 GPa and trabecular bone has an elastic modulus of approximately 3 to 11 GPa [2], [3]. Because mechanical load follows the stiffest path through a structure, the rigid implant absorbs most of the load that would otherwise pass through the surrounding bone. The bone, now receiving far less mechanical stimulation than normal, responds according to Wolff's Law and is progressively resorbed and weakened [4]. Over time this leads to bone tissue loss around the implant, loosening of the implant, and in some cases fracture of the surrounding tissue upon implant removal. Revision surgery is costly, risky, and places a significant burden on both patients and healthcare systems [5], [7].

Several approaches have been explored to reduce stress shielding. These include using lower-modulus materials such as polymer composites or titanium foam, introducing porosity into implant geometries, and modifying implant shape to improve load transfer [10], [11], [12]. Previous research has also investigated porous scaffold optimization for tissue engineering applications [11]. However, this study addresses a distinct problem: the stiffness reduction of a load bearing metallic plate using aerospace-derived topology optimization. Rather than designing a scaffold for biological growth, the objective here is to reduce excessive implant stiffness while preserving sufficient load bearing capacity.

Topology optimization is a computational method that removes unnecessary material of an object within a defined design space, given a set of loads and boundary conditions. It has been used extensively in the aerospace industry to reduce the mass of structural components such as aircraft brackets and engine mounts, where minimizing weight while maintaining structural performance is a central design goal [25]. The most established algorithm for topology optimization is the Solid Isotropic Material with Penalization method (SIMP) which was developed by Bendsoe and later refined by Sigmund [14], [15], [16].

The mathematical framework of SIMP is not specific to any one engineering domain. In aerospace, the objective is to minimize mass while maintaining structural integrity and strength. For orthopedic implants, the related but distinct goal is to reduce structural stiffness toward the range of cortical bones while maintaining sufficient mechanical integrity for load bearing. These two problems are mathematically similar, which suggests that the same computational tool can be applied to both [14], [18].

This study investigates whether aerospace-derived topology optimization can be used to design a bone fixation plate with reduced effective structural stiffness, bringing it closer to the stiffness of cortical bone and thereby reducing the effects of stress shielding. A 2D finite element model of a titanium fixation plate was built, and the SIMP algorithm was applied at five different volume fractions ranging from 30% to 70% of the full plate volume. The relative structural stiffness of each optimized design was then compared against the cortical bone stiffness range of 21 to 29 GPa [2], [3] to identify which configuration most closely matches the mechanical behaviour of natural bone under load.

The specific objectives of this study are as follows. First, implement a 2D SIMP topology optimization model of a bone fixation plate under a physiological bending load. Second, to evaluate the optimized geometry, mass reduction, and effective structural stiffness across five volume fractions. Third, to identify the volume fraction

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at which the optimized implant most closely matches cortical bone stiffness and assess the degree to which topology optimization alone can close the stiffness gap between titanium and bone.

2. Literature Review

2.1 Biological Background

Bone is composed of two main layers: an outer, rigid cortical layer and an inner, spongy trabecular bone layer. Cortical bone provides a strong and stiff framework, while trabecular bone forms a lightweight lattice-like structure that distributes load and absorbs force [1]. Trabecular bone resembles the structure of topology-optimized structures used in aerospace engineering applications. In addition to its optimized structure, bone also exhibits unique mechanical properties. Cortical bone has an elastic modulus of approximately 21 to 29 GPa whereas trabecular bone presents lower stiffness with elastic modulus from approximately 3 to 11 GPa, due to its porous and optimized architecture [2], [3]. Bone is anisotropic and heterogeneous, meaning its mechanical properties (including elastic modulus) change based on bone location in the body and loading direction.

2.2 Aerospace Engineering Background

Topology optimization is a computational method used to find the best possible distribution of material inside a given design space, based on the loads applied to it and how much material is allowed. The method was first introduced by Bendsoe and Kikuchi in 1988 through a homogenization approach, where they used mathematics to work out the ideal internal microstructure of a porous material that would maximize stiffness for a fixed amount of material [14], [18]. This was the foundation that made topology optimization a legitimate and mathematically sound engineering tool.

3. Research Method

This study used a computational approach based on the Solid Isotropic Material with Penalization (SIMP) topology optimization algorithm, implemented from scratch in Python 3.9 using only the NumPy, SciPy, and Matplotlib libraries, with no external topology optimization packages [16], [17]. The simulation was run as a parametric study across five volume fractions to find which implant geometry most closely matches the structural stiffness of cortical bone. The algorithm was split across four Python files: run_study.py, simp.py, fem.py, and filter.py, each handling a distinct part of the optimization process.

3.1 Design Domain and Finite Element Model

The bone fixation plate was modelled as a two-dimensional rectangular domain under plane-stress conditions. The domain was divided into a structured mesh of 60 x 20 bilinear quadrilateral elements (Q4), giving a 3:1 length-to-height aspect ratio that is representative of a standard cortical bone fixation plate. The mesh contained 1,200 elements and 1,302 nodes, producing 2,604 degrees of freedom in total (two translational degrees of freedom per node: horizontal displacement u and vertical displacement v).

The plane-stress assumption was used because the plate thickness is small compared to its in-plane dimensions, so through-thickness stress components can be treated as negligible. Each element stiffness matrix was calculated using a 2x2 Gauss quadrature, which integrates the bilinear shape functions of the Q4 element exactly [19]. The global stiffness matrix was assembled using a direct stiffness assembly and stored in a compressed sparse column (CSC) format for efficient solving.

3.2 Boundary Conditions and Loading

Two boundary conditions were applied to represent the clinical scenario of a bone fixation plate screwed to the cortex and loaded during weight-bearing. The left edge (x = 0) was fully fixed, meaning both translational degrees of freedom at every node on that edge were set to zero. This represents the screw anchors that attach the plate rigidly to the bone.

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A distributed downward force was applied uniformly across all nodes on the right half of the top edge (x > L/2). This represents the bending load that passes through the bone segment during activities like walking. The load was applied to only the right half of the top surface on purpose, because an asymmetric load produces a richer optimized topology that uses more of the design domain. The total force was normalized to unity across all loaded nodes.

To partially address the single-load-case limitation noted in Section 3.10, the optimization was also run at VF30 under an axial compression load case. In this case, a uniform horizontal compressive force was applied across all nodes on the right edge, with the left edge remaining fully fixed. This represents the compressive load transmitted along the bone axis during the stance phase of walking, where body weight acts parallel to the long axis of the bone. The topology produced under axial loading was compared to the bending topology at the same volume fraction to assess how sensitive the optimized geometry is to load direction.

The absolute load magnitude does not change the optimized topology because compliance scales with the square of the force, so only the relative sensitivity distribution matters for the density update [16].

3.3 Material Properties

The implant was modelled using the material properties of titanium alloy Ti-6Al-4V, which is the most used material in orthopedic implants [6]. Key parameters: Young's modulus E0 = 110 GPa [6], Poisson's ratio v = 0.3 [6], SIMP penalization exponent p = 3 [15], [16], sensitivity filter radius r min = 1.5 elements [17], maximum iterations = 100, convergence tolerance = 0.01. The minimum modulus E min was set to E0 x 10^{-9} to prevent singular stiffness matrices in voided regions [16].

3.4 SIMP Algorithm

The topology optimization problem was set up as a minimum compliance problem subject to a volume constraint:

Minimize:

$$c(\mathbf{x}) = \mathbf{u}^T \mathbf{K}(\mathbf{x}) \mathbf{u}$$

Subject to:

$$\mathbf{K}(\mathbf{x})\mathbf{u} = \mathbf{f}, \quad \frac{\sum x_e}{n_{el}} = V^*, \quad x_{min} \leq x_e \leq 1$$

Here, c is the structural compliance (a measure of flexibility), u is the nodal displacement vector, K is the global stiffness matrix, f is the external force vector, x_e is the density of element e, n_el is the total number of elements, and V* is the target volume fraction. The SIMP interpolation scheme gives each element an effective Young's modulus:

$$E(x_e) = E_{min} + x_e^p (E_0 - E_{min})$$

With p = 3, any element with an intermediate density has a modulus that is disproportionately low relative to its volume. This penalizes the optimizer for keeping grey elements and pushes the solution toward a clear solid-void topology [15], [16].

The compliance value was recorded at every iteration to monitor convergence. Figure 6.2 (presented in section 4.2) shows the compliance history for all five volume fractions on a single plot. In all cases, compliance drops sharply in the first 20 to 30 iterations as the optimizer redistributes material toward efficient load paths, then flattens as the density field approaches a stable configuration. This monotonically decreasing pattern confirms that the OC update is correctly minimizing the objective throughout the parametric study. VF50, VF60, and VF70 converged within the 100-iteration limit, reaching the tolerance at iterations 73, 67, and 84 respectively. VF30 and VF40 did not fully converge within 100 iterations, which is consistent with known SIMP behavior

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at low volume fractions where the sparse, heavily constrained topology takes longer to resolve a binary distribution [16], [24].

3.5 Sensitivity Analysis

The sensitivity of the compliance with respect to each element's density was calculated analytically at every iteration:

$$\frac{dc}{dx_e} = -p \cdot x_e^{p-1} \cdot (E_0 - E_{min}) \cdot \mathbf{u}_e^T \mathbf{k}_e^{unit} \mathbf{u}_e$$

Here, u_e is the element nodal displacement vector and k_e^{unit} is the element stiffness matrix at unit modulus. The sensitivity is always zero or negative, which confirms that adding material to any element can only reduce compliance or keep it the same [16].

3.6 Sensitivity Filter

Before each density update, the raw element sensitivities were passed through a spatial sensitivity filter. This filter was first proposed by Sigmund and Petersson [24]:

$$\frac{d\tilde{c}}{dx_e} = \frac{1}{x_e \sum_f H_{ef}} \sum_f H_{ef} \cdot x_f \cdot \frac{dc}{dx_f}$$

The weight H_ef = max(0, r_min - dist(e,f)) gives more importance to nearby elements. The filter radius was set to r_min = 1.5 elements [17]. The filter prevents checkerboard patterns and mesh-dependent results and enforces a minimum feature size of r_min [24], [17].

3.7 Density Update

Element densities were updated each iteration using the Optimality Criteria (OC) method:

$$x_e^{new} = x_e \cdot \left(\frac{-d\tilde{c}/dx_e}{\lambda} \right)^{\eta}$$

A move limit of +/- 0.2 was applied per iteration. Lambda is the Lagrange multiplier enforcing the volume constraint, and eta = 0.5 is the standard damping exponent. Lambda was found by a bisection search [16]. A lower density bound of x min = 10^-3 was used. The optimization was considered converged when the largest density of changes between iterations dropped below 0.01, or when 100 iterations were completed.

3.8 Parametric Study

The optimization was run separately at five volume fractions:

$$V^* = \{0.30, 0.40, 0.50, 0.60, 0.70\}$$

Each run started from a uniform density field at the target volume fraction, so the volume constraint was satisfied with the very first iteration. These five cases represent implants that retain between 30% and 70% of a fully solid titanium plate, with mass reductions of 70% down to 30%.

3.9 Evaluation Metrics

Three metrics were recorded for each optimized design. Mass reduction (%) was calculated as (1 - V*) x 100, relative to the fully solid plate baseline. Final compliance was the scalar value $c = u^T K(x^*) u$ at the converged density field. Effective structural stiffness (GPa) was estimated using the compliance ratio:

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$$E_{\text{eff}} = \frac{c_{\text{ref}}}{c_{\text{design}}} \times E_0$$

Here, c_ref is the compliance of the stiffest design in the study, used as a proxy for the solid plate, and E0 = 110 GPa. The values were compared against the published cortical bone stiffness range of 21 to 29 GPa [2], [3] to identify which volume fraction most closely matches natural bone [5], [7].

The E_eff is not a directly measured material modulus. Instead it compares the structural stiffness of the entire plate in response to the applied load relative to the solid titanium reference. So it measures the displacement of the entire plate (not just the material). If one wanted to know what the optimized structure's stiffness was at the material level, then homogenization calculations on the internal microstructure would be needed but are beyond the scope of this study [18]. For a relative comparison, the values of E_eff can still be compared with the cortical bone values because they describe a similar type of structural behavior dominated by bending. Read the E_eff values as a measure of how far topology optimization has pushed the plate's structural stiffness towards the bone range, not as a claim that the optimized lattice matches cortical bone as a material.

3.10 Limitations

The simulation is two-dimensional and uses plane-stress assumptions, so out-of-plane loading and through-thickness variation are not captured. The study primarily focused on a single bedngin dominated optimization case, with an additional axial compression comparison performed at VF30. However, full multi load case optimization was not implemented. Both the implant and the bone were treated as linear elastic; bone is viscoelastic and anisotropic, which this model does not account for [8], [20], [21]. The effective stiffness value E_eff is a structural estimate based on the compliance ratio, not a directly measured material modulus [18]. As explained in Section 3.9, it is used as a relative benchmark and is validated against the known titanium modulus at VF70. A proper homogenization analysis would give more accurate material-level stiffness values and is recommended for future work. No fatigue or biocompatibility analysis was carried out [12].

4. Results and Discussions

4.1 Optimized Topologies

Figures 1 to 5 show the optimized material density fields at volume fractions of 30%, 40%, 50%, 60%, and 70%. In each figure, black regions represent solid material (relative density close to 1), and white regions represent void (relative density close to 0). Grey values indicate intermediate densities that did not fully converge to binary under the SIMP penalization.

VF = 30% (Figure 1)

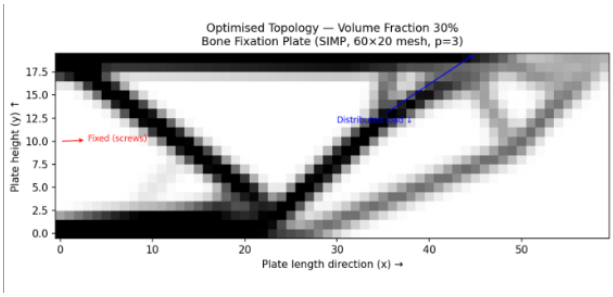


Fig. 1. *Optimized material topology at volume fraction 30%. Material is concentrated along two primary paths: top compression chord and a shear transfer strut. Black = solid titanium (density = 1), white = void (density = 0).*

The most material efficient layout is produced by the optimizer at the lowest volume fraction. The majority of the material collects in two main load paths. There is a high density top chord extending from the fixed end towards the loaded region. There is a diagonal compression strut dropping down from approximately x = 25 to the bottom center, braced by a tension chord on the bottom edge. The large central void contains little load,

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and the optimizer therefore removes it. The remainder resembles a classic cantilever truss. Here grey intermediate density areas persist, which shows that the solution has not yet converged to a binary distribution after 100 iterations. This is the behavior of SIMP at low volume fractions [16], [24].

VF = 40% (Figure 2)

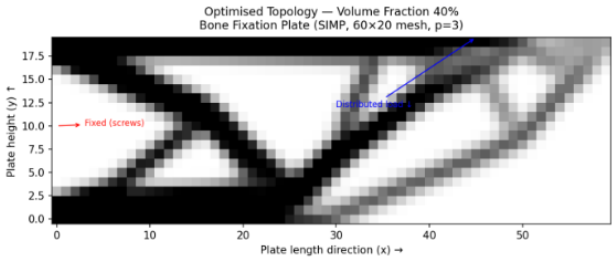


Fig. 2. *Optimized material topology at volume fraction 40%. Diagonal load transfer structure becomes more continous and additional secondary members appear in the lower left region.*

With 40% of the material retained, the topology keeps the diagonal strut pattern from VF30 but adds a secondary structural member in the lower-left part of the domain. The top chord is more continuous, and the void regions are more clearly separated. The diagonal load path from the fixed left edge toward the load zone on the right side of the top edge is more distinct, which reflects the algorithm directing material along the most efficient force transfer route. There are fewer grey regions than VF30, meaning the solution converged more cleanly.

VF = 50% (Figure 3)

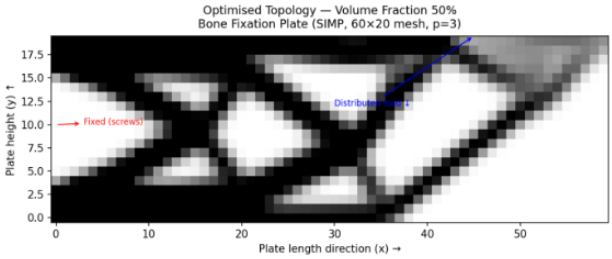


Fig. 3. *Optimized material topology at volume fraction 50%. The topology transitions from a sparse truss-like structure toward a more distributed multi-member network with several diagonal and horizontal reinforcement paths.*

The topology at 50% goes from a sparse truss to a more distributed network. Several distinct members exist now, including a top chord, a bottom chord and at least two diagonal interior members. There are large voids still present in the upper left and lower central areas. Even with half the material remaining there is a lot of plate that sees little load under this load case. The same layout can be found in aerospace structural optimization studies [25], [18].

VF = 60% (Figure 4)

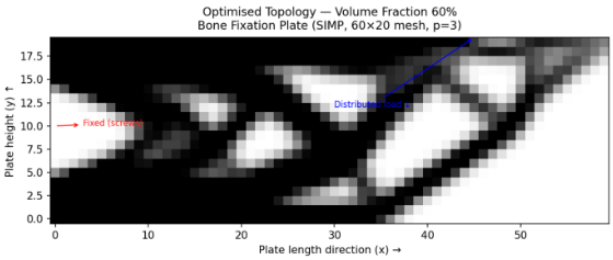


Fig. 4. *Optimized material topology at volume fraction 60%. The structure becomes increasingly dense with reduced void space and more intermiediate-density regions*

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At 60%, the topology starts to fill in, and the individual structural members are harder to distinguish. The right side of the domain, directly below the load application zone, retains higher material density. The left side has more voided regions. The higher number of grey intermediate-density elements at this volume fraction reflects the known reduction in SIMP penalization effectiveness at higher densities [16], [17].

VF = 70% (Figure 5)

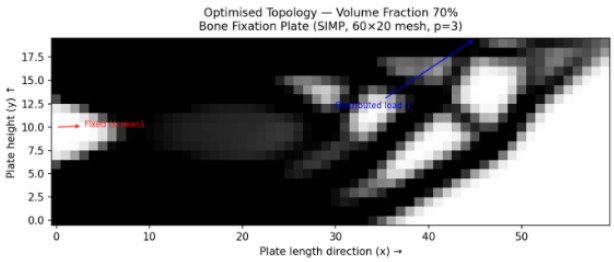


Fig. 5. *Optimized material topology at volume fraction 70%. The topology approaches a near-solid plate with selective voiding concentrated away from the primary load-transfer regions.*

This means the densest layout is at the highest volume fraction. Asymmetry due to loading only the right half of the top edge is obvious, and voiding is contained to the upper-left and lower-left. This design is almost a solid plate with selective voiding. This brings its stiffness near the 110 GPa titanium baseline [6]. All five cases show the same trend; as the volume fraction decreases, the optimizer removes material from low stress regions and adds it to efficient load paths. This agrees with what topology optimization does in aerospace structural design [25], [18] and confirms that SIMP transfers correctly to the orthopedic load case.

4.2 Compliance and Stiffness-Mass Tradeoff

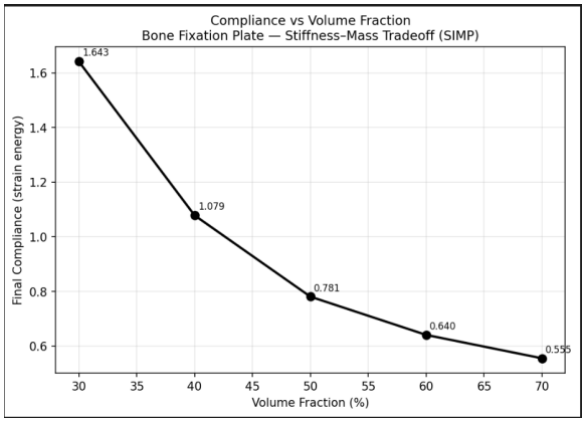


Fig. 6. *Final compliance vs volume fraction: stiffness-mass tradeoff curve.*

Figure 6 shows the final compliance at each volume fraction. The compliance curve is strongly nonlinear. Between VF30 and VF50, compliance drops steeply from 1.643 to 0.781, a reduction of 52.5%. This shows that material added in this range is very effective at increasing structural stiffness. Between VF50 and VF70, compliance only drops by a further 28.9% despite adding another 20% of material. This is a classic diminishing-returns pattern: as volume fraction increases, the remaining void regions matter less structurally, so each additional unit of material gives less stiffness in return.

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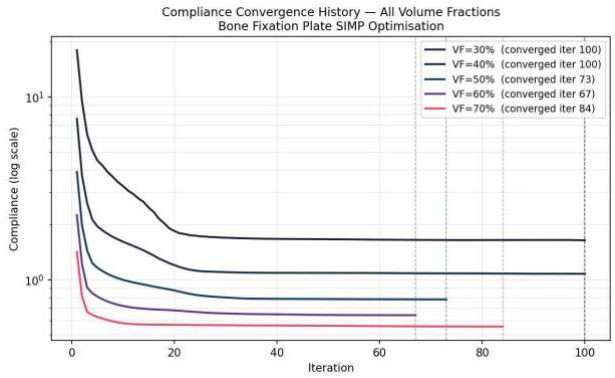


Fig 8.2. *Compliance convergence against all volume fractions, legend uses "iter" as a shortening of "iteration"*

Table 1 below summarizes the key results from all five-volume fractions:

Table. 1. *mass reduction, compliance, stiffness comparison across all volume fractions*

Volume Fraction	Mass Reduction (%)	Final Compliance	Effective Stiffness (GPa)
30%	70%	1.643	37.2
40%	60%	1.079	56.6
50%	50%	0.781	78.2
60%	40%	0.640	95.3
70%	30%	0.555	110.0

A volume fraction between 30% and 50% gives the most structurally efficient implant, achieving substantial stiffness while removing the most material. Volume fractions above 50% add significantly more mass with limited structural benefit [12], [13].

4.3 Stiffness Comparison with Cortical Bone

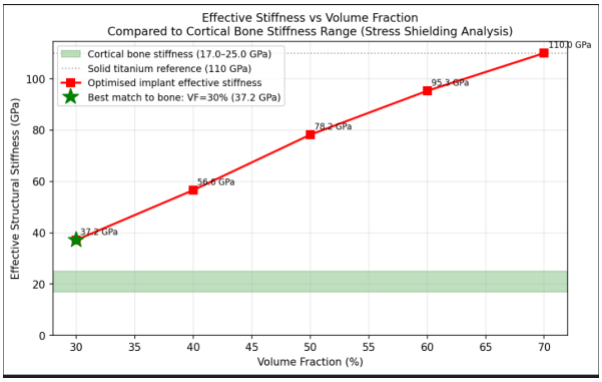


Fig. 7. *Effective structural stiffness of each optimized design vs cortical bone stiffness range of 21 to 29 GPa.*

Figure 7 compares the effective structural stiffness of each design to the range of values for cortical bone (21 to 29 GPa) [3], [8]. Section 3.9 clarifies that E eff is an estimate of structure scaled by the solid titanium baseline; it is not a material property. The five volume fractions are all outside the range for cortical bone. The volume fraction with the smallest amount of difference from cortical bone was VF30, which is 37.2 GPa, or about 28% above the upper limit and 77% above the lower limit. The effective stiffness of 110.0 GPa matches the solid titanium baseline at VF70 [6].

First, topology optimization of titanium alone is not sufficient to fully close the stiffness gap between the implant and cortical bone within the volume of fractions tested. Even at VF30, with 70% of the material removed, the effective stiffness at 37.2 GPa is still well above the bone range of 21 to 29 GPa [3]. The stiffness mismatch between metallic implants and bone is a well-documented clinical problem that has been linked to long-term bone resorption and implant loosening [5], [7].

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Second, topology optimization does achieve a large and clinically meaningful reduction in effective stiffness. Going from the solid plate at 110.0 GPa to the VF30 design at 37.2 GPa is a 66.2% reduction. The absolute stiffness gap relative to bone narrows from 81 to 89 GPa (solid plate versus bone) down to 8 to 16 GPa (VF30 versus bone). This partial reduction in stiffness mismatch is expected to reduce the severity of stress shielding in a clinical setting [7], [23], even if the bone stiffness range is not fully reached within this parametric study.

4.4 Biological Interpretation and Stress Shielding Implications

Reducing the stiffness of orthopedic implants has important biological consequences under Wolff's Law, which states that bone remodels in response to the mechanical loads it experiences [4]. Typical titanium or steel implants possess significantly greater stiffness than bone [6], [3], [8], causing the implant to absorb a higher proportion of the applied load and reducing the mechanical stimulation of the surrounding bone tissue. By partially reducing implant stiffness through topology optimization, a larger fraction of the load can be transferred into the bone, allowing for healthier bone remodeling and a reduced risk of implant loosening over time [22]. Even partial stiffness matching may mitigate stress shielding effects and support healthier bone recovery while maintaining the structural integrity required for load-bearing activities [5], [7]. These effects are particularly important in elderly or osteoporotic patients, where the surrounding bone tissue already has a compromised structure and is more vulnerable to stress-shielding induced resorption [9].

4.5 Aerospace Parallel

The structural behavior observed across the parametric study closely matches what is seen in aerospace topology optimization. In both cases, the SIMP algorithm produces truss-like load-path structures at low volume fractions, with top and bottom chords carrying axial forces and diagonal members transferring shear. This pattern is consistent with classical optimal truss theory as described in the topology optimization literature [14], [18]. The similarity confirms that the mathematical framework of aerospace topology optimization applies directly and correctly to the orthopedic problem.

The compliance versus volume fraction curve in Figure 6 also replicates the diminishing-returns behavior that has been reported in aerospace structural optimization studies [25], adding further evidence that the method behaves consistently across domains. The key difference between the two applications is the design objective: in aerospace the goal is minimum mass at a required stiffness, whereas in this study the goal is to reduce excessive implant stiffness while preserving sufficient load-bearing capability. These are mathematically related problems, and the results confirm they can be addressed with the same computational tool.

4.6 Load Case Comparison: Bending vs Axial Compression at VF30

Figure 8 superimposes the bending and axial compression layouts at VF30. They are distinct. In bending, the structure consists of a top chord and a diagonal strut running from the fixed left edge to the loaded area (see Section 4.1). Bending is characterized by axial chord loads and diagonal shear. Hence, the material accumulates in these components. In the axial configuration, the two parallel horizontal elements carry most of the load with little diagonal support. This is the shortest path for horizontal compression between the fixed and loaded ends, and represents the classical solution for axially loaded struts [14], [18].

The axial case also converged considerably faster, reaching the density-change tolerance at iteration 51, compared to 100 iterations without full convergence for the bending case at the same volume fraction. This faster convergence reflects that the axial load case produces a mechanically cleaner topology with fewer intermediate-density grey elements, because the load path under pure compression is more geometrically straightforward than under asymmetric bending.

The difference between the two topologies confirms that the optimized geometry is sensitive to load direction. An implant optimized solely for bending may not distribute axial compression efficiently, and vice versa. This supports the recommendation in Section 5 that future work should use a combined bending-plus-axial multi-load-case optimization, which would produce a single topology that performs well under both loading regimes simultaneously.

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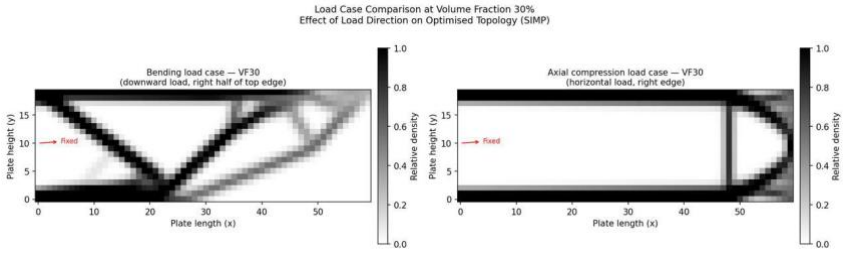


Fig. 8. *Side-by-side topology comparison at volume fraction 30%. Left: bending load case (downward load on right half of top edge). Right: axial compression load case (horizontal load on right edge)*

4.7 Mesh Convergence Study at VF30

A mesh convergence study was carried out for the VF30 optimization case to evaluate sensitivity to mesh resolution. As shown in Figure 9 below, simulations using coarser and finer meshes produced similar overall topology layouts and compliance trends, indicating that the used 60x20 mesh provided sufficient resolution for the purposes of this study.

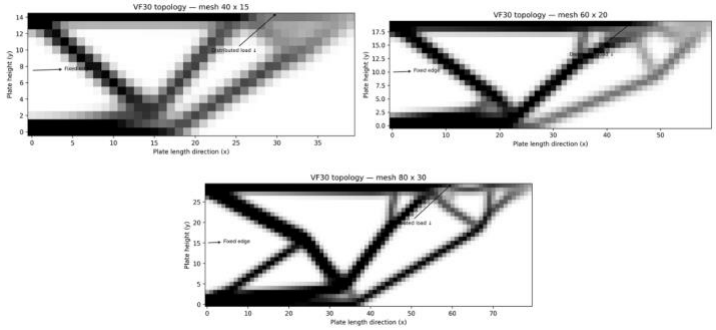


Fig. 9. *Side-by-side topology comparison at volume fraction 30%. Top left is 40x15 mesh, top right is 60x20 mesh, bottom is 80x30 mesh*

Table 2 summarizes the mesh convergence carried out for the VF30 optimization case. Three mesh resolution were tested: 40x15, 60x20 and 80x30. Although final compliance values varied, all simulations produced similar dominant load transfer paths and overall topology layouts, consisting primarily of an upper compression chord, lower tension chords and diagonal shear-transfer members. The finer meshes generated smoother and more detailed structural members, while the coarser mesh produced a more pixelated approximation of the same pattern. The optimization converged successfully for all three meshes within the specified convergence tolerance (raised to 200 iterations for this convergence study). These results indicate that the selected 60x20 mesh provides sufficient resolution to capture the principal optimization behaviour without the solution depending heavily on mesh density.

Table 2. *Shows compliance and convergence iterations for each mesh size, element count is included for completeness*

Mesh	Elements	Compliance	Convergence Iterations
40x15	600	1.415200	198
60x20	1200	1.635974	123
80x30	2400	1.032708	186

4.8 Computational Load-Response Validation

A computational load response validation was performed to verify that the optimized topologies behaved consistently under load. Each converged topology from the parametric study was re-analysed under the same bending load case and boundary conditions used during optimisation. The maximum vertical displacement was recorded for each VF and compared against the corresponding compliance values.

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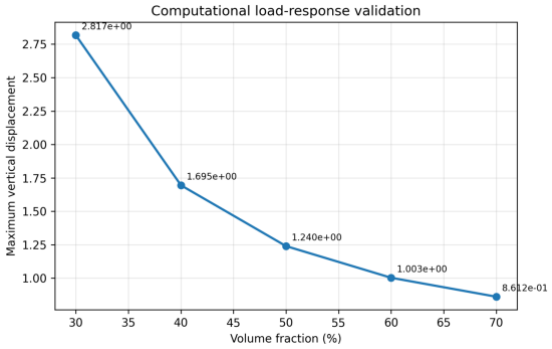


Fig. 10. *Maximum vertical displacement under identical loading conditions for each optimized topology. Lower VF structures exhibit greater deformation, consistent with their higher compliance values and lower stiffness, e+00 means 10 power 0 (1) and e-01 means 10 power –1 (0.1)*

The computational validation followed the expected mechanical trend. Lower VF topologies showed higher compliance and larger maximum vertical displacement values under identical loading conditions, while higher VF designs showed lower compliance and smaller displacements. The VF30 design produced the largest maximum vertical displacement of 2.817, whereas VF70 produced the smallest of 0.861. This behavior is consistent with classical structural mechanics, where structures with lower stiffness deform more under the same load [18], [19].

The convergence history shown in Figure 11 below supports the numerical stability of the optimization process. The compliance decreases rapidly during early iterations before approaching a stable asymptotic value, indicating successful convergence of the SIMP optimization algorithm.

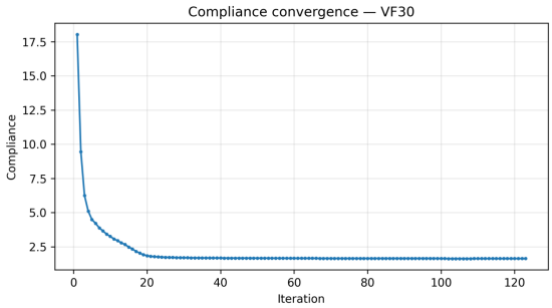


Fig. 11. *Compliance convergence history for the VF30 case. Compliance decreases rapidly during early iterations before approaching a stable asymptotic value, indicating numerical convergence of the SIMP optimization algorithm*

5. Conclusion

Topology optimization, a computational method widely used in aerospace structural design, was used in this study to investigate whether it could be applied to reduce the stiffness of a titanium implant while maintaining implant strength. The motivation behind this was the clinical problem of stress shielding, where rigid metal implants absorb most of the mechanical load that should pass through the surrounding bone, causing the bone to weaken over time under Wolff's Law, leading to bone resorption, problematic implant loosening, and costly revision surgeries [4], [5].

A 2D finite element model of a bone fixation plate was built, and the SIMP topology optimization algorithm was run at five volume fractions: 30%, 40%, 50%, 60%, and 70%. The optimized geometries, mass reductions, compliance values, and relative structural stiffness values were recorded and compared against the cortical bone stiffness range [2], [3].

Topology optimization produced a 66.2% reduction in effective structural stiffness, from 110 GPa at the solid titanium baseline down to 37.2 GPa at the lowest tested volume fraction of 30%. This was accompanied by a 70% reduction in implant mass. However, none of the five volume fractions tested produced an implant with

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effective stiffness inside the cortical bone range. The closest result was at VF30, which remained approximately 28% above the upper bound of the bone range at 37.2 GPa.

Two conclusions can be drawn from this. Topology optimization alone, within the titanium material system and the single load case used here, is not sufficient to fully close the stiffness gap between the implant and cortical bone. At the same time, the 66.2% stiffness reduction is a large and clinically meaningful result. Narrowing the absolute stiffness gap from 81 to 89 GPa down to 8 to 16 GPa is expected to meaningfully reduce the severity of stress shielding, even without reaching the bone stiffness range entirely. As discussed in Section 4.4, this partial stiffness matching is particularly beneficial for elderly and osteoporotic patients where surrounding bone is already compromised [9].

Future work should extend this approach in several directions. Running the parametric study below 30% volume fraction may identify designs that fall within the cortical bone stiffness range. A three-dimensional model with several load cases would give a more realistic picture of implant performance. Combining topology optimization with lower-modulus materials such as titanium foam or polymer composites could close the stiffness gap further [10], [11]. Finally, the optimized geometries produced in this study are well suited to additive manufacturing, which would allow physical prototypes to be produced and tested mechanically in future experiments [10].

Overall, this study demonstrates that aerospace-derived topology optimization is a viable and promising tool for reducing stress shielding in orthopedic implants. The cross-disciplinary transfer of the SIMP algorithm from aerospace structural design to orthopedic implant design produced mechanically meaningful stiffness reductions and demonstrated the potential of computational structural optimization to contribute to future orthopedic implant design strategies aimed at reducing stress shielding and improving long term implant performance.

Acknowledgements

The authors thank Andrew Stangroom of Doha College for his supervision and guidance throughout this project. Both authors contributed equally to the work. The simulation code was implemented jointly, and the manuscript was written together.

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